LENGTH CONTRACTION, RELATIVE VELOCITY AND RELATIVISTIC MOMENTUM

LEARNING GOALS

Students will:

- Know how time dilation and the constancy of the speed of light leads to length contraction.
- Know how to calculate length contraction.
- Know how relative velocities are calculated in relativistic situations.
- Know how to calculate the relativistic changes to momentum.

LENGTH CONTRACTION

If two observers are moving relative to each other, then a time dilation from one observer's point of view will be balanced by a corresponding length contraction from the other observer's point of view. The derivation of the relevant equation can be seen on pages 573-575 in your text. It is really just a matter of looking at the points of view for different observers.

$$L_m = L_s\sqrt{1-\frac{v^2}{c^2}}$$

Length contraction applies only to lengths measured parallel to the direction of the velocity. Lengths measured perpendicular to the velocity are not affected. In the equation above, L_m is the length of an object or the distance between two points according to an observer who is moving with respect to that length or distance. L_s is the same length or distance according to an observer who is not moving with respect to the length or distance. This is sometimes called the **proper length** of the object and might be designated as L_0 instead of as L_s.

The value of any measurement is tied to the frame of reference in which that measurement is taken. Differences in measurements of length (parallel to the motion) and time will occur between one frame of reference and another. Normally these differences are too small to be observed, but as the relative speeds approach the speed of light, these differences become more significant.

GAMMA

To simplify problems involving relativistic speeds, physicists have assigned the symbol γ (gamma in the Greek alphabet) to the value of

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

Therefore, time dilation and length contraction become

$$\Delta t_m = \gamma \Delta t_s \qquad \text{and} \qquad L_m = \frac{L_s}{\gamma}$$

Calculations involving dilated time and contracted lengths imply that v is always less than c.

PRACTICE PROBLEMS

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RELATIVE VELOCITY

If an observer on earth sees two space ships approaching each other and she measures the speed of ship 1 wrt (with respect to) herself to be 0.8c to the right and ship 2 to be 0.7c to the left. An observer in ship 1 also sees the approach of ship 2. What velocity does he see ship 2 moving at wrt his ship 1?

In our lesson on relative velocities we would solve this by writing a vector equation like:

$$\vec{v}_{21} = \vec{v}_{2g} + \vec{v}_{g1}$$

where the velocity of ship 2 wrt ship 1 is the velocity of ship 2 wrt the ground and the velocity of the ground wrt ship 1. This represents the non-relativistic velocity.

Calculate the non-relativistic result as a multiple of c.

$\vec{v}_{21} = -0.7c - 0.8c =$ **$-1.5c$ (left)** + is right

But according to Einstein, nothing can exceed the speed of light.

The relativistic version of the above equation would require an extra adjustment term. The derivation gives:

$$\vec{v}_{21} = \frac{\vec{v}_{2g} + \vec{v}_{g1}}{1 + \frac{\vec{v}_{2g}\,\vec{v}_{g1}}{c^2}} \quad \therefore \quad \vec{v}_{21} = \frac{-0.7c - 0.8c}{1 - (-0.7)(0.8)} = -0.962c \text{ (left)}$$

PRACTICE PROBLEM

Observer on earth sees a ship travelling at 0.5c [fwd] wrt the ground. A plasma cannon fires a pulse out the front of the ship so that the ground observer sees the pulse travel at 0.9c [fwd] wrt the ground. How fast do the observers in the ship see the pulse travel wrt the ship?

+ is forward

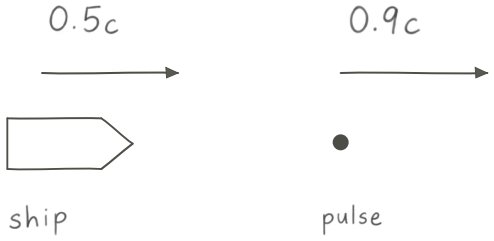
$v_{pg} = 0.9c, \quad v_{sg} = 0.5c$

$$v_{ps} = \frac{v_{pg} - v_{sg}}{1 - \frac{v_{pg}\,v_{sg}}{c^2}}$$

$$v_{ps} = \frac{0.9c - 0.5c}{1 - (0.9)(0.5)}$$

$$= \frac{0.4c}{0.55}$$

$v_{ps} =$ **$0.73c$ [fwd]**



Earth frame

p = pulse, s = ship, g = ground

PRACTICE PROBLEMS

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1. A Martian leaves Mars in a spaceship that is heading to Venus. On the way, the spaceship passes earth with a speed v=0.80c relative to it. Assume that the three planets do not move relative to each other during the trip. The distance between Mars and Venus is 1.20×10^{11} m, as measured by a person on earth.
   (a) What does the Martian measure for the distance between Mars and Venus?
   (b) What is the time of the trip (in seconds) as measured by the Martian?

2. Two spaceships A and B are exploring a new planet. Relative to this planet, spaceship A has a speed of 0.60c, and spaceship B has a speed of 0.80c. What is the ratio D_A/D_B of the values for the planet's diameter that each spaceship measures in a direction that is parallel to its motion?

3. At what speed is the magnitude of the relativistic momentum of a particle three times the magnitude of the nonrelativistic momentum?

4. A woman is 1.6 m tall and has a mass of 55 kg. She moves past an observer with the direction of the motion parallel to her height. The observer measures her relativistic momentum to have a magnitude of 2.0×10^{10} kg m/s. What does the observer measure for her height?

5. You are driving down a two-lane country road, and a truck in the opposite lane is traveling toward you. Suppose that the speed of light in a vacuum is c = 65 m/s. Determine the speed of the truck relative to you when
   (a) your speed is 25 m/s and the truck's speed is 35 m/s and
   (b) your speed is 5.0 m/s and the truck's speed is 55 m/s.
   The speeds given in parts (a) and (b) are relative to the ground.

6. A spacecraft approaching the earth launches an exploration vehicle. After the launch, an observer on earth sees the spacecraft approaching at a speed of 0.50c and the exploration vehicle approaching at a speed of 0.70c. What is the speed of the exploration vehicle relative to the spaceship?

7. Spaceships of the future may be powered by ion-propulsion engines in which ions are ejected from the back of the ship to drive it forward. In one such engine the ions are to be ejected with a speed of 0.80c relative to the spaceship. The spaceship is traveling away from the earth at a speed of 0.70c relative to the earth. What is the velocity of the ions relative to the earth? Assume that the direction in which the spaceship is traveling is the positive direction, and be sure to assign the correct plus or minus signs to the velocities.

8. The spaceship Enterprise 1 is moving directly away from earth at a velocity that an earth-based observer measures to be 0.65c. A sister ship, Enterprise 2, is ahead of Enterprise 1and is also moving directly away from earth along the same line. The velocity of Enterprise 2 relative to Enterprise 1 is 0.31c. What is the velocity of Enterprise 2, as measured by the earth-based observer?

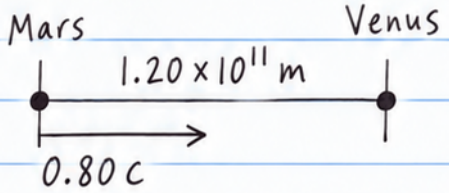
Answers: 1. a) 7.2×10^{10} m b) 300 s 2. 1.3 3. 0.94c 4. 1.0 m
5. a) 50 m/s b) 56 m/s 6. 0.31c 7. -0.23c 8. 0.80c

Length contraction / relative velocity

1) a) planets are at rest in Earth's frame

$L_s = 1.20 \times 10^{11}\ m, \quad v = 0.80\,c$

$$L_m = L_s\sqrt{1 - \frac{v^2}{c^2}}$$



$$L_m = (1.20\times10^{11})\sqrt{1 - 0.80^2}$$

$$L_m = 7.2 \times 10^{10}\ m$$

b) $$\Delta t_M = \frac{L_m}{v}$$

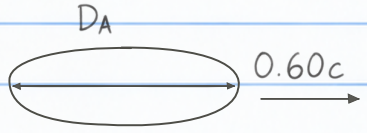
$$= \frac{7.2 \times 10^{10}}{0.80\,(3.00\times10^{8})}$$

$$\Delta t_M = 300\ s$$

2) $$D_A = D_s\sqrt{1 - 0.60^2} = 0.80\,D_s$$

$$D_B = D_s\sqrt{1 - 0.80^2} = 0.60\,D_s$$

$$\frac{D_A}{D_B} = \frac{0.80}{0.60}$$





$$\frac{D_A}{D_B} = 1.3$$

slower ship measures the larger diameter

Relativity (continued)

3) $p_{rel} = \gamma m v$, $p_{classical} = mv$

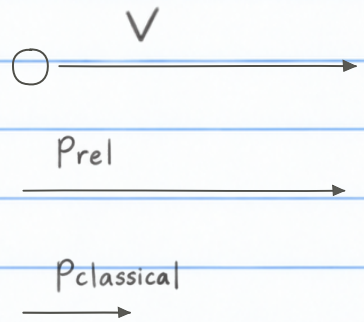
$$\frac{p_{rel}}{p_{classical}} = \gamma = 3$$

$$3 = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$1 - \frac{v^2}{c^2} = \frac{1}{9}$$

$$v = c\sqrt{1 - \frac{1}{9}}$$

$$\boxed{v = 0.94c}$$



4) $p = \gamma m v$

$$\frac{p^2}{m^2c^2} = \frac{\gamma^2 v^2}{c^2} = \gamma^2 - 1$$

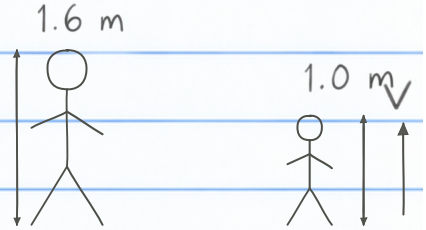
$$\gamma = \sqrt{1 + \left(\frac{p}{mc}\right)^2}$$

$$\gamma = \sqrt{1 + \left(\frac{2.0 \times 10^{10}}{55(3.00 \times 10^8)}\right)^2}$$

$$\gamma = 1.571$$

$$L_m = \frac{L_s}{\gamma} = \frac{1.6}{1.571}$$

$$\boxed{L_m = 1.0\ m}$$



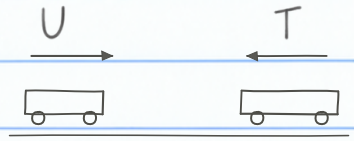
only height contracts since motion is parallel to her height

Relativity (continued)

5) + is your direction, $c = 65$ m/s

T = truck, U = you, G = ground

$$v_{TU} = \frac{v_{TG} - v_{UG}}{1 - (v_{TG} v_{UG})/c^2}$$



a) $$v_{TU} = \frac{-35 - 25}{1 - (-35)(25)/65^2}$$

$$= -49.7 \text{ m/s}$$

speed = 50 m/s toward you

b) $$v_{TU} = \frac{-55 - 5.0}{1 - (-55)(5.0)/65^2}$$

$$= -56.3 \text{ m/s}$$

speed = 56 m/s toward you

6) + is toward Earth

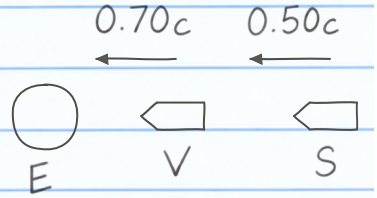
V = vehicle, S = spacecraft

$v_{VE} = 0.70c$, $v_{SE} = 0.50c$

$$v_{VS} = \frac{v_{VE} - v_{SE}}{1 - (v_{VE} v_{SE})/c^2}$$

$$= \frac{0.70c - 0.50c}{1 - (0.70)(0.50)}$$

$$= 0.20c/0.65$$



$v_{VS} = 0.31c$ toward Earth

Relativity (continued)

7) + is away from Earth

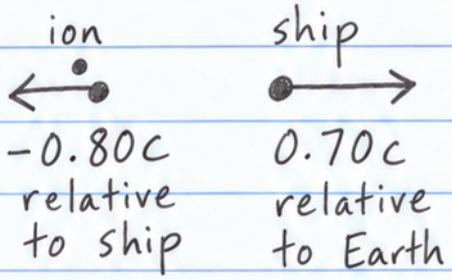
$v_{IS} = -0.80c$, $v_{SE} = +0.70c$

$$v_{IE} = \frac{v_{IS} + v_{SE}}{1 + v_{IS} v_{SE}/c^2}$$

$$= \frac{-0.80c + 0.70c}{1 + (-0.80)(0.70)}$$

$$= \frac{-0.10c}{0.44}$$

$$v_{IE} = -0.23c$$



ions move toward Earth

8) + is away from Earth

$v_{21} = 0.31c$, $v_{1E} = 0.65c$

$$v_{2E} = \frac{v_{21} + v_{1E}}{1 + v_{21} v_{1E}/c^2}$$

$$= \frac{0.31c + 0.65c}{1 + (0.31)(0.65)}$$

$$= \frac{0.96c}{1.2015}$$

$$v_{2E} = 0.80c \text{ away from Earth}$$

